



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8518**

AV:	2025-26	Session:	08-10-2025	Date:
Sem:	1	Slot:	10:00am - 11:00am	C Code:
Exam Type:	Written	Block:	401	CSA13
Course Section:	1	Seat No:	2023601002	

L_29_526_986_L_1505_38317



Name of the candidate. Arpita Bori

Candidate's UID number:

2	0	2	3	6	0	1	0	0	2
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER: 7

PROGRAM: CE - AJML

DAY AND DATE: Wednesday 08-10-2025

COURSE NAME: Deep learning

~~I / II / III / IV / V / VI / VII / VIII~~

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	(1)	
	(b)	<u>Bias-variance decomposition:</u>
		1] Bias is the difference of the actual and the predicted out when the model is been trained.
		2] variance is the difference of actual and the predicted value when the trained model is applied on the testing dataset.
		3] 70-30 or 80-20 split is used to divide the dataset into training and testing set.
		4] Bias-variance decomposition is useful when dealing with the error (difference between the actual & the predicted value.)
		5] Expected square error of the predictor $MSE = \sum_{i=1}^n (y - \hat{y})^2$
		6] <u>Overfitting</u> : It occur when the model learns the pattern and data in the training set so well that it performs well on the training set but fails to perform well on the testing set.
		This happens when the model overlearns the training data and cannot learn the patterns in the testing dataset
		7] variance and bias play a crucial role to decide whether the model is overfitted or underfitted.
		8] Low Bias & High variance results in overfitting
		9] the low bias while the generalising the errors the bias and the variance is calculated to calculate the difference of actual & predicted values.

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Question
No.

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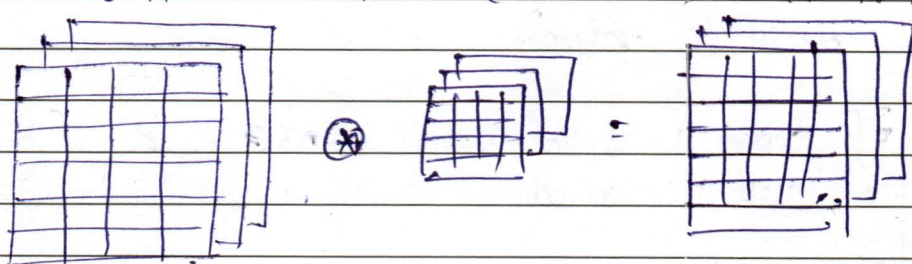
(2)

(a) 1] for the standard 2D convolution the convolutional operations are carried out like. Suppose the input is of $N \times N$ and filter is of $F \times F$.

then the output size will be $(N-F+1) \times (N-F+1)$

2] for the depth wise separable convolution - meaning that the input has several layers

i.e. $F \times F \times 3$ $\otimes$ $F \times F \times 3$
 $\swarrow \searrow$ $\swarrow \searrow$
 R G B R G B

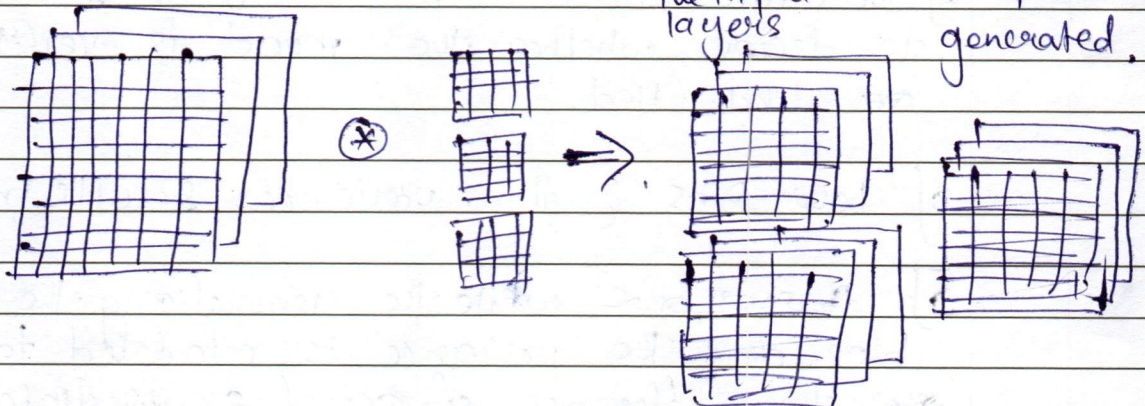
It will result in $(F-N+1) \times (F-N+1) \times 3$


3] Now for input size $H \times W \times C$ - Input
filter = $F \times F \times K$

$$H \times W \times C \otimes F \times F \times K = (H-F+1) \times (W-F+1) \times C \times K$$

$$= [(H-F+1) \times (W-F+1) \times C] \times K$$

Suppose this are the input layers
No. of outputs generated.





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So in the given example let's suppose the

$H = N_1$ $W = N_2$ $F = \text{filter}$

$C = \text{layers}$



assume as RGB (3)

$K = \text{No of filter } 3$

So it will generate output of 3 ~~different~~ images
will the provide layers.

4] Depthwise separable convolutional is important to perform the ~~a~~ convolutional operation on the different ~~a~~ layers of the input

This way the convolutional operation is applied separately on every layer. which helps in retaining those layer further.

Example $\Rightarrow$ Separable convolutions are applied on the colored image (layers $\rightarrow R, G, B$).

5] Theoretical justification is ~~ad~~ given above on how the layers are used in the separable convolutional.

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(2)

(b) 1] undercomplete encoder:

AutoEncoder are something which consists of 3 main components.

1] Encoder $\rightarrow$ which encodes the data into and performs the transformation.
 $z = f_{\theta}(x)$.

2] latent code (z) = encoded data.

3] decoder $\rightarrow$ which decodes the latent code.
 $\hat{x} = f_{\theta}(z)$

• undercomplete encoders compresses the original input into compress transformation which only holds the salient information of the input.

• This is further used to reconstruct the original input.

$$d(z) < d(x).$$

$\rightarrow$ This is used for the dimensionality reduction.

4] Now when we train it with MSE, which is the mean squared error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

it calculates the loss with the use of the MSE.

5] This is equivalently used by PCA in the linearly separable data for dimensionality reduction.

6] while This autoencoder works for non-linear data.



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(2)

(6)

7] Though the data types might be different but the function to calculate the loss is similar.

8] So the subspace is equivalent to PCA. (Principal component Analysis).

9] Further, the overcomplete autoencoder without the regularization doesn't compress the input, instead it is equal or greater than the input.

i.e. $\dim(Z) > \dim(X)$.

10] Hence, an overcomplete autoencoder learns the identity function, where the latent code stores the identity of the input with the dimensions as it has more room.

ii] It is useful for generative tools.

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(3)

(a) i] In RNN the recursive gradient expression

$$\frac{\partial L}{\partial h_{t-k}}$$

$$RNN \rightarrow h_t = \sigma(w_h [h_{t-1}, x_t] + b)$$

here the h_{t-1} is the previous hidden state and x_t is the current input.

So if the equation is derived it will look something like,

$$h_t = \sigma(w_h [h_{t-1}, L_t] + b)$$

2] But this has a problem where the hidden state is update at step which overrides the previous value.

3] Resulting in loss of hidden state which was calculated way before.

for this Long Short Term Memory comes into play.

$$4] \text{Forget gate in LSTM} = f_t = \sigma(w_f [h_{t-1}, x_t] + b)$$

5] forget gate in LSTM is used to forget the data which was stored and is not relevant in the current state.

6] This way only the information is stored in LSTM for a particular period of time and then it is ~~stored~~ forgotten if any new information needs to be stored.

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(3)
(a)

7] So the forget gate not only forgets the previous or ~~less~~ previously stored information but also restores the new information for the further use.

8] This way it prevents the information to be completely forget which in turns prevents the vanishing gradients under suitable initialization.

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(3)

(b) GRU (Gate Recurrent unit).

1] It is similar to LSTM which helps in storing the previous information and updating it with new.

So the information is ~~not~~ ~~lose~~ is not-loss.

2] Though it is the ~~an~~ alternative of LSTM the gates are different.

a] update gate $\rightarrow$ used to update the information.

$$z_t = \sigma(w_z [h_{t-1}, x_t])$$

b] Reset gate $\rightarrow$ used to forget the information

$$r_t = \sigma(w_r [h_{t-1}, x_t])$$

c] output gate $\rightarrow o_t = \sigma(w_o [h_{t-1}, x_t])$.
So above were the gating equations of GRU.

3] for updating the hidden state it first calculate the candidate by the equation.

$$\hat{h}_t = \tanh(w_h [r_t * h_{t-1}, x_t])$$

4] Now the hidden state is calculated with

$$h_t = z_t \cdot (1 - \hat{h}_{t-1}) + z_t \cdot \hat{h}_t$$

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(3)

(b)

5] GRU has the fewer parameters as compared to LSTM because of the fewer gates used in the GRU.

6] while LSTM has forget gate, input gate, output gate, candidate, calculating the hidden state with the additional equation.

7] Applications with Natural language process, vocabularies.

finding grammatical errors in the sentence.

Image recognition.

8] here are the fields where GRU can be useful than LSTM.

Sr. No.

8518



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[illegible]

[illegible]

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